Half-Weyl regularity from the curvature equation

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1. A differential budget supplied by the equation

Let (M, g) be a closed oriented smooth four-manifold. Regard $W = W^+$ as a section of the metric bundle $E = \mathrm{Sym}_0(\Lambda^+)$, with its induced connection and Hilbert–Schmidt norm. Define the residual of the half-Weyl equation by

$$Z = \nabla^*\nabla W + \frac{s}{2}W - 6W^2 + 2|W|^2 I, \qquad q = \fint_M |Z|^2. \tag{1}$$

The rough Laplacian is nonnegative. All averages use the original metric.

Theorem 1 (equation-to-budget estimate). Suppose

$$|\operatorname{Ric}| \leqslant K, \qquad |W^+| \leqslant \Lambda \quad \text{on } M. \tag{2}$$

Then

$$\boxed{\fint_M \left(|\nabla^2 W^+|^2 + |\nabla W^+|^4\right) \leqslant C_{K,\Lambda}(1+q).} \tag{3}$$

Let the ordered eigenvalues be $\lambda_1 \leqslant \lambda_2 \leqslant \lambda_3$. If also

$$\lambda_2 - \lambda_1 \geqslant \rho > 0, \tag{4}$$

the smooth orthogonal projector P onto the lowest eigenline defines the global endomorphism

$$A = \frac{\lambda_1}{2}(3P - I). \tag{5}$$

It satisfies

$$\boxed{H_2(A) := \fint_M |\nabla^2 A|^2 \leqslant C_{K,\Lambda,\rho}(1+q).} \tag{6}$$

The constants are independent of volume, injectivity radius, diameter, Sobolev constants and W^-. No derivative of the curvature of the bundle E is assumed or estimated.

In particular, if $\mathrm{Ric} = -g$, then $K = 2$, $s = -4$, and $Z = 0$. Thus an upper bound on the *one-sided* Weyl tensor and the displayed eigenline gap supply the Hessian budget automatically:

$$H_2(A) \leqslant C_{\Lambda,\rho}. \tag{7}$$

The same zero-residual conclusion applies to harmonic self-dual Weyl curvature, $\delta W^+ = 0$, under (2). In that case the classical half-Weyl Weitzenböck equation gives $Z = 0$, even without the Einstein equation.

The residual version is useful when the equation is not exactly harmonic. It does not assert that an L^∞ bound on Ricci alone controls q; the latter is an explicit differential input.

2. The bundle Hessian argument

We give the argument because applying a full-curvature estimate to a four-tensor would introduce an unnecessary W^- hypothesis.

Let z be a section of a metric bundle with connection curvature F_{ij}, and write

$$H_{ij} = \nabla_i \nabla_j z, \qquad L = \operatorname{tr} H.$$

The Hessian includes the cotangent Levi-Civita correction. Use the full two-index convention $|F|^2 = \sum_{ij} |F_{ij}|^2_{\mathrm{HS}}$. First interchange the Hessian slots, using $H_{ij} - H_{ji} = F_{ij} z$. Then integrate by parts and commute once on the bundle-valued one-form ∇z. One obtains

$$\begin{aligned} \int |H|^2 = {} & \int |L|^2 - \int \mathrm{Ric}_{ij} \langle \nabla_i z, \nabla_j z \rangle \\ & + \sum_{ij} \int \langle F_{ij} \nabla_j z, \nabla_i z \rangle + \frac{1}{2} \sum_{ij} \int |F_{ij} z|^2. \end{aligned} \tag{8}$$

In detail, the difference between the left side and $\sum_{ij} \int \langle H_{ij}, H_{ji} \rangle$ is exactly the last term in (8). In the mixed expression, the commutator acts on ∇z, not on Fz. Its cotangent contribution contracts to Ricci curvature; its bundle contribution is the third term in (8). The remaining trace term integrates to $\int |L|^2$. This order avoids differentiating F.

For completeness, integration of $\operatorname{div}(|\nabla z|^2 \langle z, \nabla z \rangle^\sharp)$ gives the dimension-four interpolation estimate

$$\left(\int |\nabla z|^4 \right)^{1/2} \leqslant 4 \|z\|_\infty \left(\int |\nabla^2 z|^2 \right)^{1/2}. \tag{9}$$

Indeed the differentiated gradient norm contributes at most $2\|z\|_\infty |H| |\nabla z|^2$, and the trace Hessian contributes at most $2\|z\|_\infty |H| |\nabla z|^2$. Cauchy–Schwarz proves (9). This is integration by parts, not a Sobolev inequality.

Assume $\|z\|_\infty \leqslant B$, $|\operatorname{Ric}| \leqslant K$, and $|F| \leqslant F_0$. Equation (8) and Cauchy–Schwarz imply

$$⨍ |\nabla^2 z|^2 \leqslant ⨍ |\nabla^* \nabla z|^2 + (K + C F_0) ⨍ |\nabla z|^2 + \tfrac{1}{2} F_0^2 B^2. \tag{10}$$

Also

$$⨍ |\nabla z|^2 = ⨍ \langle z, \nabla^* \nabla z \rangle \leqslant B \left(⨍ |\nabla^* \nabla z|^2 \right)^{1/2}. \tag{11}$$

Consequently, if the rough-Laplacian squared average is at most $C_0(1+q)$, then (10), followed by (9), bounds the Hessian squared average and fourth-power gradient average by $C(B, K, F_0, C_0)(1+q)$. Every term before averaging is proportional to volume. No lower volume bound is introduced.

3. Applying the argument to the half-Weyl equation

The curvature operator of an oriented four-manifold has blocks

$$\mathscr{R} = \begin{pmatrix} W^+ + sI/12 & \mathrm{Ric}^\circ \\ \mathrm{Ric}^\circ & W^- + sI/12 \end{pmatrix}, \tag{12}$$

where the off-diagonal entries use the usual identification with trace-free Ricci curvature. The action of $\mathfrak{so}(4) = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$ on Λ^+ is trivial on the second summand. Therefore

$$|F^{\Lambda^+}| \leqslant C(|W^+| + |s| + |\mathrm{Ric}^\circ|), \qquad |F^E| \leqslant C |F^{\Lambda^+}|. \tag{13}$$

The second estimate is the induced commutator action on endomorphisms. It is important that W is treated as an E-valued section. The W^- block does not occur in (13).

Since $|s| \leqslant 2|\operatorname{Ric}|$, (1)–(2) give

$$\nabla^*\nabla W = Z - \tfrac{s}{2}W + 6W^2 - 2|W|^2 I, \qquad \fint |\nabla^*\nabla W|^2 \leqslant C_{K,\Lambda}(1+q). \tag{14}$$

The identity part has Hilbert–Schmidt norm $\sqrt{3}$. Now (13) and (2) bound the bundle curvature; (10)–(11) apply with $z = W$ and $B = \Lambda$. Finally (9) proves (3).

For an Einstein metric, contracted Bianchi gives $\delta W^+ = 0$. The harmonic-Weyl Weitzenböck equation is precisely $Z = 0$. For $\operatorname{Ric} = -g$ its sign-normalized form is

$$\nabla^*\nabla W = 2W + 6W^2 - 2|W|^2 I. \tag{15}$$

The negative Kähler–Einstein eigenvalues $(-2/3, 1/3, 1/3)$ solve its algebraic right side, as they must.

4. Differentiating the spectral model

Put $\lambda = \lambda_1$. On the finite-dimensional space of real symmetric matrices, a simple eigenvalue and its rank-one projector depend smoothly on the matrix. This requires no separation between λ_2 and λ_3.

For clarity, let

$$S = (W - \lambda I)^{-1}(I - P),$$

where the inverse is taken on the orthogonal complement of P. For a symmetric variation K_1, differentiation of $WP = \lambda P$ and $P^2 = P$ gives

$$D\lambda[K_1] = \operatorname{tr}(PK_1), \qquad DP[K_1] = -SK_1P - PK_1S. \tag{16}$$

Since $\|S\|_{\mathrm{op}} \leqslant \rho^{-1}$, a further differentiation gives the uniform bounds

$$\begin{aligned} |D\lambda[K_1]| &\leqslant |K_1|, & |D^2\lambda[K_1,K_2]| &\leqslant C\rho^{-1}|K_1||K_2|, \\ |DP[K_1]| &\leqslant C\rho^{-1}|K_1|, & |D^2P[K_1,K_2]| &\leqslant C\rho^{-2}|K_1||K_2|. \end{aligned} \tag{17}$$

One may obtain the last estimate without differentiating an inverse on a moving subspace: use the Riesz formula for P on a circle of radius $\rho/2$ around the eigenvalue at the base matrix. Resolvents on this fixed circle have norm at most $2/\rho$. Each derivative inserts one additional resolvent, yielding (17). This is a local derivative formula; a single spectral contour for every point of M is not required.

The map in (5) is orthogonally equivariant, so its Euclidean derivatives give covariant chain rules in a normal orthonormal frame. In particular

$$\begin{aligned} |\nabla A| &\leqslant C(1 + \Lambda/\rho)|\nabla W|, \\ |\nabla^2 A| &\leqslant C(1 + \Lambda/\rho)|\nabla^2 W| + C(\rho^{-1} + \Lambda\rho^{-2})|\nabla W|^2. \end{aligned} \tag{18}$$

Squaring, averaging, and applying (3) proves (6). The same reasoning also proves

$$\fint \left(|\nabla^2 P|^2 + |\nabla P|^4 + |\nabla^2\lambda|^2 + |d\lambda|^4\right) \leqslant C_{K,\Lambda,\rho}(1+q). \tag{19}$$

This is a regularity statement for the entire isolated eigenline, not an estimate depending on a choice of local eigensection.

5. Composition with negative-amplitude selection

Suppose (2) and (4) hold, and set

$$r = \fint |\operatorname{Ric} + g|^2, \qquad e = \frac{1}{2} \fint (\lambda_3 - \lambda_2)^2, \qquad d = r + e. \tag{20}$$

Trace-freeness and (4) imply

$$0 = \lambda_1 + \lambda_2 + \lambda_3 \geqslant 3\lambda_1 + 2\rho, \qquad -\lambda_1 \geqslant 2\rho/3. \tag{21}$$

On the orientation cover of the lowest real eigenline, choose the smooth normalized form ω with $|\omega|^2 = 2$. Set $t = -3\lambda_1/4$. Then

$$\rho/2 \leqslant t \leqslant 3\Lambda/4, \quad P = \tfrac{1}{2}\omega \otimes \omega, \quad A = t(2I/3 - \omega \otimes \omega), \quad |W - A|^2 = \tfrac{1}{2}(\lambda_3 - \lambda_2)^2. \tag{22}$$

All squared densities below descend to M, and averages on the cover equal averages on the base.

Corollary 2 (sourced curvature-spectral application). Under (2) and (4), the variable-amplitude selection theorem gives

$$\boxed{\fint \left[|dt|^2 + (t - \tfrac{1}{2})^2 + |\nabla\omega|^2 \right] \leqslant C_{K,\Lambda,\rho}\left(d + \sqrt{d(1+q)} \right).} \tag{23}$$

The right side now uses the residual of an equation for actual curvature. There is no separately prescribed $H_2(A)$ hypothesis. This is a direct composition of (6) with Theorem 1 of `line1_variable_negative_selection.md`; no new monopole assumption is used.

In the exact negative-Einstein case, put $\epsilon = (\fint (\lambda_3 - \lambda_2)^2)^{1/2}$. Then $q = r = 0$, $e = \epsilon^2/2$, and $\epsilon \leqslant 2\Lambda$, so (23) implies

$$\fint \left[|d\lambda_1|^2 + (\lambda_1 + \tfrac{2}{3})^2 + |\nabla\omega|^2 \right] \leqslant C_{\Lambda,\rho}\epsilon. \tag{24}$$

It also follows that

$$\fint \left[|\nabla W^+|^2 + \operatorname{dist}\left(W^+, \mathscr{K}_-\right)^2 \right] \leqslant C_{\Lambda,\rho}\epsilon, \tag{25}$$

where $\mathscr{K}_-$ is the orthogonal orbit in $\operatorname{Sym}_0(\mathbb{R}^3)$ of $\operatorname{diag}(-2/3, 1/3, 1/3)$. To prove the derivative part, write $R = W - A$. Equations (3), (6) bound the averaged squared rough Laplacian of R. Integration by parts yields

$$\fint |\nabla R|^2 \leqslant \left(\fint |R|^2 \right)^{1/2} \left(\fint |\nabla^*\nabla R|^2 \right)^{1/2} \leqslant C_{\Lambda,\rho}\epsilon.$$

The derivative of A is controlled by dt and $\nabla\omega$, hence by (24). For the zeroth-order part, with $h = \lambda_3 - \lambda_2$, diagonalization gives the exact formula

$$\operatorname{dist}(W, \mathscr{K}_-)^2 = \tfrac{8}{3}(t - \tfrac{1}{2})^2 + \tfrac{1}{2}h^2. \tag{26}$$

The regularity theorem is useful beyond this particular composition. In the exactly harmonic regime, a separate weighted first-order identity can improve (24) for the lowest-eigenline branch. Equations (3), (6), and (19) still supply second-order control and apply to a nonzero equation residual; no claim of optimality is made for (24).

6. Mathematical scope and sources

The new step here is the removal of an auxiliary spectral-model derivative budget by the actual half-Weyl equation, with constants uniform under collapse and without controlling the opposite Weyl tensor. Its proof combines standard bundle calculus and spectral functional calculus with the previously established variable-amplitude selection estimate. It is not a new classification of exact Einstein metrics, nor a claim that pointwise Ricci pinching supplies bounded W^+, the eigenline gap, or the source in (1).

The classical half-Weyl equation is stated in LeBrun, *Einstein Metrics, Harmonic Forms, and Conformally Kähler Geometry*, arXiv:1903.00956v2, Section 2, equation (9). Its weighted version reduces to (1) with $Z = 0$ when the weight is one. The curvature block decomposition and its endomorphism conventions appear in Section 1, equation (3). Primary source.

The classical zero-splitting conformal-Kähler mechanism is Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*, Compositio Math. 49 (1983), 405–433, Proposition 5(i). Primary source.

The local proof inputs are the bounded-form criterion, Section 2; the variable calibration; and the negative-amplitude selection theorem. The argument above reproduces the bundle estimate to make its curvature dependence explicit. The accompanying finite checker checks the algebraic equation, spectral projection derivatives, residual norm, and target distance. It does not replace the integration-by-parts proof or verify publication priority.